The Retention Advantage Ratio: A Time-Indexed Framework for Evaluating Distributed Practice Efficiency

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Abstract. Effect sizes in practice-distribution research are usually reported at a single retention interval, even though the advantage of distributed over massed or blocked practice changes with the delay to test. This paper treats the *retention advantage ratio* $\mathcal{R}(t) = P_{\text{distributed}}(t)/P_{\text{blocked}}(t)$ as the primary object of analysis and works with its logarithm. Under a power law of forgetting with a shared time scale, the log-ratio separates exactly into two terms: an initial-strength term that is constant in time, and a decay-rate term that grows linearly in $\ln(1 + \alpha t)$. The decomposition yields the crossover delay in closed form and makes the decay-rate difference $\Delta\lambda$ the slope of a straight line. Applied to the descriptive retention surface fitted by Cepeda, Vul, Rohrer, Wixted, and Pashler (2008), the framework gives model values $\mathcal{R} = 1.04$, 1.34, and 1.73 at 7, 35, and 70 days, against data-implied values of 1.10, 1.59, and 2.11. At 350 days the model ratio (5.16) is roughly three times the data-implied ratio (1.77), because the model's massed-practice recall is near the floor; long-delay ratios computed from fitted surfaces should therefore not be reported as findings. I review interleaving and contextual-interference evidence that shows the same acquisition-retention dissociation, note evidence that cuts against a simple generalization, and propose a reporting standard: log-ratios with standard errors at specified retention intervals, and $\Delta\lambda$ when three or more intervals are available.

Keywords: spacing effect, retention advantage ratio, log response ratio, desirable difficulties, retention interval, practice distribution

1. Introduction

The question "Which practice schedule is more efficient?" hides a dependence on time. Performance during or shortly after practice often favors massed or blocked schedules, while delayed tests favor distributed or interleaved schedules, a dissociation that Soderstrom and Bjork (2015) describe as the distinction between learning and performance. A single effect size therefore answers the question only for the delay at which it was measured, yet the field has no standard way to express how much the advantage changes with retention interval.

Existing formal work addresses related but distinct questions. Cepeda et al. (2008) fit a retention surface, final recall as a function of inter-study gap and retention interval (RI), and traced the "ridgeline" of optimal gaps. That answers *which* gap is best at each delay, not *how much better* it is than a benchmark. Activation-based and context-based models (Pavlik & Anderson, 2005; Mozer, Pashler, Cepeda, Lindsey, & Vul, 2009) predict absolute recall for each schedule, from which a ratio could be computed but is not treated as a construct. Paas and van Merriënboer's (1993) efficiency measure combines performance and mental effort at a single test. Kim, Wong-Kee-You, Wiseheart, and Rosenbaum (2019) analyzed workplace-training data from 10,514 learners with a generalized linear mixed model and found a spacing × retention-interval interaction, with the optimal spacing interval increasing as the retention interval increased, but they did not isolate the growth of the relative advantage as a named function.

This paper proposes the *retention advantage ratio*

$$\mathcal{R}(t) = \frac{P_{\text{distributed}}(t)}{P_{\text{blocked}}(t)} \tag{1}$$

where $P(t)$ is the probability of correct recall at retention interval t, and analyzes it on the log scale, $\ell(t) = \ln \mathcal{R}(t)$, which is the log response ratio familiar from meta-analysis (Hedges, Gurevitch, & Curtis, 1999). The contributions are: (1) an exact decomposition of $\ell(t)$ into an initial-strength term and a decay-rate term; (2) an analytical application to the Cepeda et al. (2008) surface, including where that application fails; and (3) a reporting standard built on $\ell(t)$ at specified intervals.

2. Mathematical Framework

2.1 The Power Law of Forgetting

Retention after practice is well described by a power function of time (Wixted & Ebbesen, 1991; Wixted, 2004):

$$P(t) = m(1 + \alpha t)^{-\lambda} \tag{2}$$

where m is recall at $t = 0$, α is a temporal scaling parameter, and λ is the decay exponent. If spacing slows forgetting, it does so by lowering λ.

2.2 Log-Ratio Decomposition

Suppose distributed (d) and blocked (b) conditions share α but may differ in m and λ. Then

$$\ell(t) = \ln \mathcal{R}(t) = \underbrace{\ln \frac{m_d}{m_b}}_{\text{initial strength}} + \underbrace{\Delta\lambda \, \ln(1 + \alpha t)}_{\text{decay rate}} \tag{3}$$

with $\Delta\lambda = \lambda_b - \lambda_d$. Equation 3 is exact, not an approximation. Plotted against $x = \ln(1 + \alpha t)$, the log-ratio is a straight line: the intercept is the initial-strength difference and the slope is $\Delta\lambda$. Three consequences follow directly.

Monotonicity. $d\ell/dt = \Delta\lambda\,\alpha/(1 + \alpha t)$, so $\mathcal{R}(t)$ increases with t if and only if $\Delta\lambda > 0$. Under the pure power law with no lower asymptote it increases without bound; Section 3.5 shows why that limit should not be taken literally.

Crossover. If $m_d < m_b$ and $\Delta\lambda > 0$, the two conditions are equal at

$$t^* = \frac{1}{\alpha}\left[\left(\frac{m_b}{m_d}\right)^{1/\Delta\lambda} - 1\right] \tag{4}$$

Blocked practice is ahead for $t < t^*$ and distributed practice for $t > t^*$. In words, the crossover is the delay at which the accumulated decay-rate benefit repays the initial-strength cost. Figure 1 illustrates the shape with parameters chosen for illustration only.

Estimability. With observations at several intervals, $\Delta\lambda$ is the slope of a regression of ℓ on $\ln(1 + \alpha t)$, which gives a direct test of whether a manipulation changes the rate of forgetting rather than only the level.

2.3 Cost-Adjusted Variant

When conditions differ in the amount of practice needed to reach criterion, a cost-adjusted ratio is

$$\mathcal{R}_c(t) = \frac{n^{(b)}}{n^{(d)}}\,\mathcal{R}(t), \qquad \ell_c(t) = \ell(t) + \ln \frac{n^{(b)}}{n^{(d)}} \tag{5}$$

where n is a measure of practice cost (trials, presentations, or time). On the log scale the cost term simply adds. Most experiments equate practice, in which case $\mathcal{R}_c = \mathcal{R}$; where they do not, the direction of the cost term matters (Section 4.3).

3. Application to a Published Retention Surface

3.1 The Cepeda et al. (2008) Surface

Cepeda et al. (2008) taught 1,354 participants 32 obscure facts, gave a review session after a gap of 0 to 105 days, and tested after a retention interval of 7, 35, 70, or 350 days, for 26 gap × RI conditions. The "zero" gap was about 3 minutes. They reported a function that fits final recall with $R^2 = .98$:

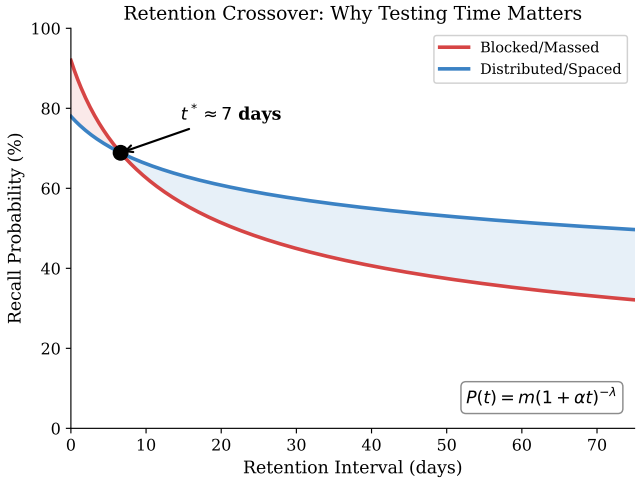


Figure 1: Illustrative retention curves for blocked (red) and distributed (blue) practice under $P(t) = m(1 + \alpha t)^{-\lambda}$, with shared $\alpha = 0.15$; blocked $m = 0.92$, $\lambda = 0.42$; distributed $m = 0.78$, $\lambda = 0.18$. Equation 4 gives $t^* \approx 6.6$ days. Parameters are illustrative and not fitted to any dataset.

$$\text{recall}(t, g) = A(g)\,(bt + 1)^{-R(g)} \tag{6}$$

$$A(g) = p + (1 - p)e^{-ag} \tag{7}$$

$$R(g) = 1 + c[\ln(g + 1) - d]^2 \tag{8}$$

with $b = 0.011$, $p = 0.760$, $a = 0.017$, $c = 0.092$, and $d = 3.453$. Here A is recall at $t = 0$, which declines from 1 at $g = 0$ toward the asymptote p as the gap grows, and R is a U-shaped rate of forgetting with its minimum $R = 1$ at $g = e^d - 1 \approx 30.6$ days. The time scale b is shared by all gaps, which is exactly the shared-α condition of Equation 3.

Two cautions govern everything that follows. First, the authors state that several other functions fit the data about as well and that they do not claim this one is uniquely accurate, only reasonable. Second, the surface was fitted to RIs from 7 to 350 days and gaps from 0 to 105 days; values outside that range are extrapolations.

3.2 Decomposition Within the Surface

For any fixed gap g, since $A(0) = 1$,

$$\ell(t; g) = \ln A(g) + [R(0) - R(g)]\ln(1 + bt) \tag{9}$$

which is Equation 3 with $m_d/m_b = A(g)$ and $\Delta\lambda = R(0) - R(g)$, where $R(0) = 1 + c\,d^2 = 2.097$. The intercept $\ln A(g)$ is never positive, so every gap starts at a disadvantage under this surface, and the slope is positive for all gaps up to about $e^{2d} - 1 \approx 1{,}000$ days. Figure 2

shows the lines for four tested gaps, and Table 1 gives the implied crossover delays from Equation 4. Longer gaps start lower and climb faster. The crossovers for short gaps fall below the shortest tested RI of 7 days, so they are properties of the fitted function rather than observations.

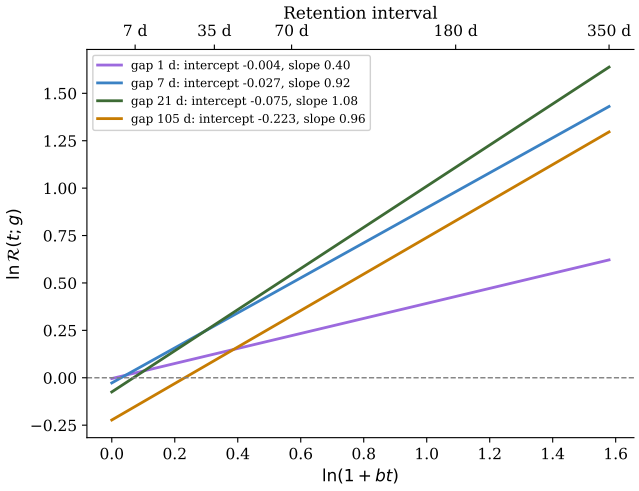


Figure 2: Log-ratio decomposition of the Cepeda et al. (2008) surface (Eq. 9). Each fixed gap is a straight line in $\ln(1+bt)$ with intercept $\ln A(g)$ (initial-strength cost) and slope $R(0)-R(g)$ (decay-rate benefit). The top axis shows the corresponding retention intervals.

Table 1: Initial-strength cost, decay-rate benefit, and crossover delay by gap, from the Cepeda et al. (2008) surface

Gap (days)	$A(g)$	$R(g)$	$R(0)-R(g)$	t^* (days)
1	0.996	1.701	0.396	0.9
7	0.973	1.174	0.923	2.7
21	0.928	1.012	1.085	6.5
30.6	0.903	1.000	1.097	8.9
105	0.800	1.135	0.962	23.7

3.3 Optimal-Gap Ratio

The headline quantity compares the best gap at each delay with massed practice:

$$\mathcal{R}(t) = \frac{\max_g \text{recall}(t, g)}{\text{recall}(t, 0)} \tag{10}$$

Because each fixed-gap ratio increases with t, so does their maximum (Appendix A). Table 2 reports $\mathcal{R}(t)$ together with its two components at the optimal gap g_t^*; the components add to $\ell(t)$ up to rounding. Figure 3 shows recall as a function of gap, and Figure 4 shows $\mathcal{R}(t)$.

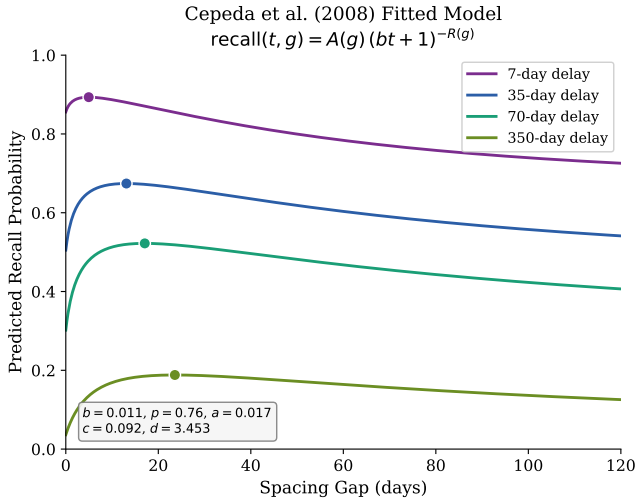


Figure 3: Recall predicted by the Cepeda et al. (2008) surface (Eqs. 6–8) as a function of gap at four retention intervals. Circles mark the optimal gap at each delay. As the delay grows, the optimal gap moves right and massed recall falls faster.

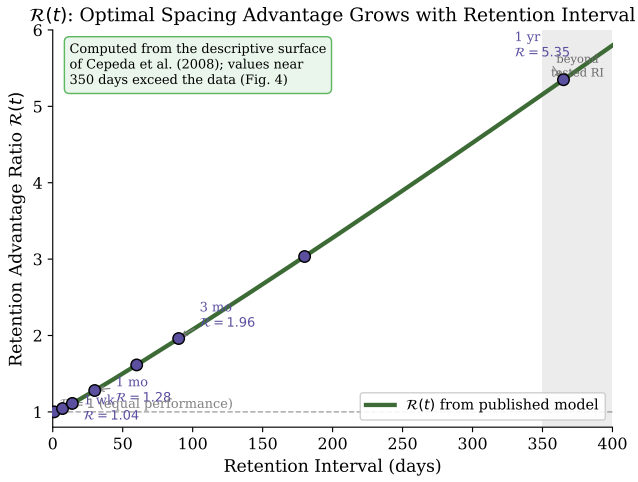


Figure 4: Optimal-gap retention advantage $\mathcal{R}(t)$ computed from the Cepeda et al. (2008) surface. The shaded band lies beyond the longest tested RI. The steep rise at long delays is driven by a massed-recall denominator approaching zero and is not supported by the data at 350 days (Fig. 5).

Table 2: Optimal-gap retention advantage computed from the Cepeda et al. (2008) surface, with log-ratio decomposition at the optimal gap

Retention interval	g_t^* (days)	Massed recall	Optimal recall	$\mathcal{R}(t)$	$\ln A(g_t^*)$	$[R(0) - R(g_t^*)]\ln(1+bt)$	$\ell(t)$
1 day[a]	0.5	0.977	0.978	1.00	−0.002	0.003	0.001
7 days	5.0	0.856	0.893	1.04	−0.020	0.062	0.043
14 days	8.0	0.741	0.823	1.11	−0.031	0.136	0.105
30 days	12.2	0.550	0.704	1.28	−0.046	0.293	0.247
35 days	13.1	0.505	0.674	1.34	−0.049	0.338	0.289
70 days	17.1	0.302	0.522	1.73	−0.062	0.610	0.548
90 days	18.4	0.236	0.463	1.96	−0.067	0.740	0.673
180 days	21.5	0.101	0.307	3.03	−0.076	1.186	1.110
350 days[b]	23.6	0.036	0.188	5.16	−0.083	1.723	1.640

Computed from the published parameters with continuous optimization over $g \in [0, 500]$. [a]Below the shortest tested RI (7 days); extrapolated. [b]The data-implied value at this RI is 1.77 (Section 3.4); the model value should not be read as an empirical estimate.

3.4 Model Versus Data

Cepeda et al. (2008) reported that the best tested gap improved recall over the zero gap by 10%, 59%, 111%, and 77% at RIs of 7, 35, 70, and 350 days, with best tested gaps of 1, 11, 21, and 21 days. These imply $\mathcal{R} =$ 1.10, 1.59, 2.11, and 1.77 ($\ell = 0.10$, 0.46, 0.75, and 0.57). The model gives 1.04, 1.34, 1.73, and 5.16 (Figure 5).

Through 70 days, model and data agree on the direction and approximate rate of growth, with the data running somewhat higher. Part of that gap is expected: the data-implied value takes the best of six or seven noisy cells at each RI, which biases it upward, while the model smooths across cells.

At 350 days the two disagree sharply. The data-implied ratio falls from 2.11 to 1.77, while the model ratio rises to 5.16. The reason is the denominator. The model's massed recall at 350 days is 0.036, so a small absolute error in that one cell produces a large error in the ratio. A surface that fits recall proportions with $R^2 = .98$ can still misrepresent ratios badly wherever the reference condition is near the floor. This is a general property of ratio measures, not a defect specific to this surface, and it is the main reason the reporting standard in Section 8 asks for observed log-ratios with standard errors rather than ratios read off fitted curves.

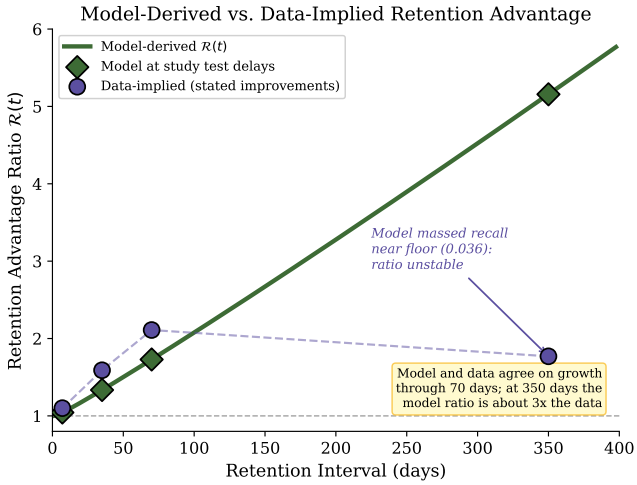


Figure 5: Model-derived $\mathcal{R}(t)$ (line and diamonds) versus data-implied values from the improvements reported by Cepeda et al. (2008) (circles). Both rise through 70 days. At 350 days the model ratio is about three times the data-implied ratio because the model's massed recall is near the floor.

3.5 Floors and the Long-Run Limit

The pure power law decays toward zero, so under it $\mathcal{R}(t) \to \infty$ whenever $\Delta\lambda > 0$. Long-term studies argue against taking that limit seriously. Bahrick (1984) found that a substantial portion of Spanish vocabulary learned in school remained accessible for decades, which he called permastore. A retention function with a floor π,

$$P(t) = \pi + (m - \pi)(1 + \alpha t)^{-\lambda} \tag{11}$$

implies $\mathcal{R}(t) \to \pi_d/\pi_b$ as $t \to \infty$: the ratio levels off at the ratio of the floors instead of growing without bound.

Bahrick and Phelps (1987) give the most direct numbers. Eight years after relearning Spanish vocabulary to

criterion, recall was 15% for a 30-day intersession interval, 8% for a 1-day interval, and 6% for a zero interval (sessions run back to back), against 1% for untrained controls. The 30-day/zero-interval ratio is 2.5, which the authors themselves report. If those values approximate the floors, the long-run ratio for that design is of order 2.5, which is consistent with the data-implied value of 1.77 at 350 days and inconsistent with the model's 5.16. The comparison is indirect: the two studies used different materials, training regimes, and definitions of massed practice, so the Bahrick and Phelps values bound plausibility rather than calibrate the Cepeda surface.

4. Meta-Analytic Context

4.1 Spacing Effect Sizes

Table 3 lists two widely cited aggregate estimates. Donovan and Radosevich (1999) found large differences by task type and by intertrial interval: effects were largest when spaced practice involved very short intertrial intervals and smallest for intervals over a day. Aggregates of this kind illustrate the problem this paper addresses: they average over retention intervals at which the advantage differs.

Table 3: Aggregate effect sizes for distributed practice

Source	d	Precision
Donovan & Radosevich (1999)[a]	0.46	95% CI [0.42, 0.50]
Donoghue & Hattie (2021)[b]	0.85	$SE = 0.053$

[a]63 studies, 112 effect sizes; task-cluster means ranged from 0.07 to 0.97. [b]Distributed practice within a meta-analysis of ten learning techniques (242 studies, 1,619 effects).

4.2 Optimal Gap Relative to Retention Interval

Cepeda et al. (2008) found that the optimal gap was about 20% of the RI for delays of a few weeks and about 5% at one year. The fitted surface gives optimal gaps of 5.0, 13.1, 17.1, and 23.6 days at 7, 35, 70, and 350 days, or 71%, 37%, 24%, and 7% of the RI. The surface's 7-day value is well above the best tested gap of 1 day, another reminder that the function is a smooth summary rather than a cell-by-cell description.

4.3 Long-Term Retention and Cost

The Bahrick and Phelps (1987) design also allows a cost adjustment, because training continued to criterion and the conditions needed different amounts of practice. The average number of presentations per word pair was 3.0 for the zero interval and 5.5 for the 30-day interval, so the spaced condition was the more expensive one. Equation 5 gives

$$\mathcal{R}_c(8\text{ yr}) = \frac{3.0}{5.5} \times 2.5 \approx 1.36, \qquad \ell_c = 0.92 - 0.61 = 0.31.$$

Spacing still wins after accounting for its extra practice, but by much less than the raw ratio suggests. The number of training sessions differed only slightly across intervals (means of 7.3 to 7.8).

Bahrick, Bahrick, Bahrick, and Bahrick (1993) compared spaced schedules with each other: 13 sessions at 56-day intervals yielded retention comparable to 26 sessions at 14-day intervals. Relative to the 14-day schedule, the 56-day schedule therefore has $\mathcal{R} \approx 1$ at half the number of sessions, or $\mathcal{R}_c \approx 2$ with sessions as the cost unit. Because each session was trained to criterion, sessions are an imperfect cost unit; as Cepeda et al. (2008) note, this design confounds total study time with gap.

5. Extensions: Interleaving and Contextual Interference

5.1 Interleaving

Brunmair and Richter (2019) meta-analyzed 238 effect sizes from 59 studies and found an overall interleaving benefit of $g = 0.42$, 95% CI [0.34, 0.50], strongly moderated by material (Table 4).

Table 4: Interleaving effects by material type (Brunmair & Richter, 2019)

Material	Hedges' g	95% CI
Paintings	0.67	[0.57, 0.77]
Naturalistic photographs	0.35	[0.16, 0.55]
Mathematical tasks	0.34	[0.11, 0.57]
Artificial pictures	0.31	[0.18, 0.44]
Expository texts	0.21	[−0.06, 0.47]
Words	−0.39	[−0.64, −0.14]

For words, blocking was better, so $\mathcal{R} < 1$ and the framework's growth prediction does not apply. More importantly for this paper, Brunmair and Richter found no evidence that retention interval moderated the interleaving effect. Their meta-regression compares different studies at different delays, which is a weak test of within-study growth, but it is evidence against assuming that interleaving advantages grow with delay the way spacing advantages do.

Within-study evidence is scarce. In a classroom study of seventh-grade mathematics, Rohrer, Dedrick, and Stershic (2015) found test scores of 80% versus 64% (interleaved versus blocked) after 1 day and 74% versus 42% after 30 days:

$$\mathcal{R}(1\text{ d}) = \tfrac{0.80}{0.64} = 1.25, \qquad \mathcal{R}(30\text{ d}) = \tfrac{0.74}{0.42} = 1.76 \quad (12)$$

or $\ell = 0.22$ and 0.57. The growth is consistent with a decay-rate benefit, but it comes from one study with two time points, and, as in most classroom interleaving designs, interleaving also spaced practice of each problem type, so the two manipulations are not fully separable.

5.2 Contextual Interference in Motor Learning

Random practice schedules in motor learning show the same dissociation. In Shea and Morgan's (1979) study, mean total time (seconds; lower is better) was:

Schedule	Acquisition	Retention
Blocked	1.32	1.73
Random	1.69	1.31

Blocked practice was faster during acquisition and slower at retention. Two recent meta-analyses by Czyż and colleagues quantify the effect across studies. For retention, Czyż, Wójcik, Solarská, and Kiper (2024) reported a medium overall benefit of random practice (three-level model SMD = 0.63, 95% CI [0.33, 0.93], total $I^2 = 90\%$), large in laboratory settings (0.92) and small and non-significant in applied settings (0.23, 95% CI [−0.16, 0.62]); by age, the effect was negligible in young participants (0.02), medium in adults (0.63), and large in older adults (1.45). For transfer, Czyż, Wójcik, and Solarská (2024) reported SMD = 0.55, 95% CI [0.25, 0.86], total $I^2 = 86\%$, again larger in laboratory (0.75) than applied settings (0.37, 95% CI [−0.02, 0.69]), with a negligible effect in young participants (0.12). These are single delayed-test estimates, so they establish the sign of $\mathcal{R}$ at retention but not its growth.

5.3 Variability of Practice

Variable practice, in which learners practise several versions of a task, is often grouped with spacing and interleaving. Schema theory (Schmidt, 1975) predicts benefits for generalization, and Kerr and Booth (1978) found that children who practised tossing at two distances outperformed children who practised only at the criterion distance when tested at that criterion distance. I did not identify meta-analytic evidence that measures variable-practice effects at several retention intervals, so this paper makes no quantitative claim about $\mathcal{R}(t)$ for variability.

6. Theoretical Basis

The new theory of disuse (Bjork & Bjork, 1992) distinguishes storage strength from retrieval strength and holds that the gain in storage strength from a study or retrieval event is larger when current retrieval strength is lower. A compact way to state this, as my formalization rather than the authors' notation, is

$$\Delta S_s \propto f(S_r), \quad f'(S_r) < 0 \tag{13}$$

Massed practice occurs while retrieval strength is high, so it buys little durable storage; spacing lets retrieval strength fall before the next event. Pyc and Rawson (2009) tested the related retrieval effort hypothesis by manipulating the interval between practice trials and the criterion level. Difficult but successful retrievals produced better final test performance than easier successful retrievals, and higher criterion levels produced diminishing returns. In the language of Equation 3, these mechanisms predict a positive $\Delta\lambda$: they act on the rate of forgetting, not only on its starting level.

7. Generality Hypothesis

Hypothesis. For a manipulation that lowers acquisition performance and raises delayed performance relative to a control, $\ell(t)$ increases with t over educationally relevant intervals, that is, $\Delta\lambda > 0$.

The hypothesis predicts that multi-interval studies of interleaving (in domains where it helps) and of random practice will show a positive slope of ℓ on $\ln(1+\alpha t)$. The Rohrer et al. (2015) data are consistent with it; the absence of retention-interval moderation in Brunmair and Richter (2019) is a reason for doubt.

Shared time scale. Equation 3 requires a common α. For spacing the Cepeda et al. (2008) surface builds this in. For interleaving it is less secure, because interleaving changes encoding as well as timing, for example by promoting discrimination between categories (Kornell & Bjork, 2008; Birnbaum, Kornell, Bjork, & Bjork, 2013). If $\alpha_d \neq \alpha_b$,

$$\mathcal{R}(t) = \frac{m_d}{m_b} \cdot \frac{(1+\alpha_d t)^{-\lambda_d}}{(1+\alpha_b t)^{-\lambda_b}} \tag{14}$$

and $\ell(t)$ need not be linear in $\ln(1+\alpha t)$ or even monotonic. Variability manipulations might likewise alter context-dependent retrieval (Smith & Vela, 2001) in ways that affect time scale rather than decay rate. Testing for a common time scale requires at least three retention intervals per condition.

8. Proposed Reporting Standard

1. Log-ratio at specified intervals. Report $\ell(t) = \ln(\bar{P}_d/\bar{P}_b)$ at each retention interval tested, with the interval stated, and with its approximate sampling variance (Hedges et al., 1999):

$$\widehat{\mathrm{Var}}(\ell) = \frac{s_d^2}{n_d \bar{P}_d^2} + \frac{s_b^2}{n_b \bar{P}_b^2} \tag{15}$$

Report the absolute difference $\bar{P}_d - \bar{P}_b$ alongside it, because ratios become unstable when the reference condition is near the floor (Section 3.4). At minimum, include one test within a day of practice and one delayed test; 30 days is a practical delayed interval for semester-length courses.

2. Decay-rate difference. With three or more intervals, fit Equation 2 by condition, or regress ℓ on $\ln(1+\alpha t)$, and report $\Delta\lambda$ with its uncertainty. With exactly two intervals and an assumed α, Equation 3 gives

$$\Delta\hat{\lambda} = \frac{\ell(t_2) - \ell(t_1)}{\ln[(1+\alpha t_2)/(1+\alpha t_1)]} \tag{16}$$

which should be labeled as a two-point estimate that depends on the borrowed α (for example $b = 0.011$ from Cepeda et al., 2008).

3. Cost. Where practice was not equated, report the practice each condition received so that ℓ_c can be computed (Eq. 5).

Studies that report only ℓ at two standard intervals can still be pooled across laboratories to estimate the slope at the meta-analytic level, which no single small study can do. Table 5 collects the values available now.

Table 5: Available retention advantage values

Comparison	Source	RI	$\mathcal{R}$
Spaced vs. massed	Cepeda (data)	7 d	1.10
		35 d	1.59
		70 d	2.11
		350 d	1.77
	Cepeda (model)	7 d	1.04
		35 d	1.34
		70 d	1.73
	Bahrick & Phelps	8 yr	2.5 (1.36[a])
Interleaved vs. blocked	Rohrer et al.	1 d	1.25
		30 d	1.76

[a]Cost-adjusted for presentations per item. The model value at 350 days (5.16) is omitted because it conflicts with the data.

9. Limitations

Descriptive surface. The analytical results inherit the Cepeda et al. (2008) function, which its authors describe as one reasonable fit among several. Properties such as the crossover delays in Table 1 may not survive a different functional form.

Extrapolation and floors. Values below 7 days or above 350 days are extrapolations, and the model's ratio at 350 days is contradicted by the data. Ratios computed where the reference condition is near the floor are unreliable however good the overall fit.

Single calibration. The analytical curve comes from one study of trivia facts. Generalization to other materials, learners, and tasks is untested.

Optimal versus typical spacing. $\mathcal{R}(t)$ in Equation 10 uses the best gap, which learners rarely achieve, and data-implied values select the best of several noisy cells. Both are upper bounds on what a typical schedule delivers.

Shared time scale. The linear decomposition requires $\alpha_d = \alpha_b$, which is built into the spacing surface but untested for interleaving and variability.

Domain boundaries. The framework's growth prediction applies only where the manipulation helps at delay; it does not apply to interleaving of words (Brunmair & Richter, 2019), and contextual-interference benefits are small or absent in applied settings and in young learners (Czyż et al., 2024a, 2024b).

Cost data. Efficiency in the full sense requires practice-cost data, which most studies do not report.

10. Conclusion

The advantage of distributed practice is a function of retention interval, and the log of the retention advantage ratio is the natural scale on which to see it. Under a power law with a shared time scale, the log-ratio is the sum of a constant initial-strength term and a decay-rate term that grows linearly in $\ln(1 + \alpha t)$, which gives the crossover in closed form and turns $\Delta\lambda$ into a slope. Applied to the Cepeda et al. (2008) surface, the framework reproduces the growth seen in the data through 70 days and fails at 350 days, where the model's massed recall is near the floor. That failure is the practical lesson: report observed log-ratios with standard errors at stated intervals, and estimate $\Delta\lambda$ from multi-interval data, rather than reading long-delay ratios off fitted curves. Whether interleaving and contextual interference share the decay-rate signature of spacing is an open empirical question that the proposed standard is designed to answer.

Code Availability

All computed values and figures are produced from the published Cepeda et al. (2008) parameters by `compute_Rt_analytical.py` and `generate_figures.py`, and checked by `test_model.py`, at https://github.com/mike8685/retention-advantage-ratio, archived on Zenodo (doi:10.5281/zenodo.23484598).

Changes from Version 1.0.1

Reframed the analysis around the log-ratio decomposition (Eqs. 3, 9); corrected the sign of the exponent in the crossover formula; removed the headline one-year value from the abstract and qualified all long-delay model values; removed an unsupported claim about raw massed recall at 350 days; corrected the Bahrick and Phelps (1987) cost adjustment, which previously combined two studies and ran in the wrong direction; replaced an incorrect citation (Kaplan et al., 2019) with Kim et al. (2019); replaced a single merged citation with the two Czyż et al. (2024) contextual-interference meta-analyses and moved them from the variability section; corrected the Brunmair and Richter (2019) category labels and reported their null retention-interval moderation; removed unsourced crossover ranges, an unsourced Cepeda et al. (2006) effect size, and an unsourced row of variability estimates; corrected confidence intervals for Donovan and Radosevich (1999) and Donoghue and Hattie (2021); corrected the Mozer et al. (2009) author list; corrected DOIs for Pyc and Rawson (2009) and Rohrer et al. (2015), which pointed to unrelated articles; added missing references (Bahrick, 1984; Hedges et al., 1999; Smith & Vela, 2001).

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A. Monotonicity of the Optimal-Gap Ratio

For a fixed gap g, Equation 9 gives

$$\frac{d\ell(t;g)}{dt} = \frac{b\,[R(0) - R(g)]}{1 + bt} \tag{17}$$

which is positive whenever $R(g) < R(0)$. Since $R(g) < R(0)$ exactly when $|\ln(g+1) - d| < d$, this holds for $0 < g < e^{2d} - 1 \approx 1{,}000$ days, a range that contains every maximizer considered here.

Let $\mathcal{R}(t) = \max_g \mathcal{R}(t;g)$ and take $t_1 < t_2$ with maximizer g_1 at t_1. Then

$$\mathcal{R}(t_2) \geq \mathcal{R}(t_2;g_1) > \mathcal{R}(t_1;g_1) = \mathcal{R}(t_1),$$

so $\mathcal{R}(t)$ is strictly increasing. The argument needs no assumption about the shape of recall in g; it uses only that each fixed-gap ratio increases with t. The result is a property of the fitted surface and, as Section 3.4 shows, does not by itself establish that observed ratios keep growing at long delays.